Supporting information to

YR-MPE: a modular desktop platform for agent-operable and reproducible phylogenetic workflows

YR-MPE Case Studies

Worked examples for molecular phylogenetics and evolution analysis

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Chapter 1

Single-gene maximum-likelihood phylogeny

1.1 Question

Can a single non-coding marker recover the relationships among closely related species when the marker is variable, contains indels, and includes sequences of unknown orientation?

1.2 Data

The example follows the ITS2 data set of Wu and Liu (2019): 25 sequences from the nuclear ITS2 region, spanning *Euconulus*, *Vitrea*, *Oxychilus*, *Sinoxychilus* and the outgroup *Pseudiberus*. The GenBank accessions are MK299689–MK299747, AY014113–AY014116, JF837183 and MN056414–MN056417. The sequences are unaligned and include the 5.8S/28S flanks; the tutorial analyses the ITS2 region between them.

1.3 Before you start

Create a persistent workspace. ITS2 is non-coding, so use MAFFT, MUSCLE 5 or Clustal Omega; MACSE is for protein-coding sequences and is not appropriate here.

1.4 Steps

1. Import `its2_raw.fasta` with **ALIGN → Open Sequence Files**. Confirm that 25 sequences appear in the workspace.
2. Open **ALIGN → Align by → MAFFT**. In **Advanced Parameters**, set **Direction adjustment** to `--adjustdirection`; the ITS2 records come from mixed sources and need not share an orientation. Run MAFFT and import the alignment.

3. Trim with **ALIGN → Trim Alignment by → TrimAl**, mode `-automated1`. The example alignment contracts from 1,458 to 1,408 columns; inspect the preview before importing.

4. Optionally run **DISTANCE → Calculate & Plot Substitution Saturation** with pairwise deletion. ITS2 is variable but should not show the near-complete transition/transversion overlap of a saturated marker.

5. Run **MODEL → Find Best Substitution Model (ML)** with BIC. The example selects `TIM2+F+G4`.

6. Run **PHYLOGENY → Maximum Likelihood → IQ-TREE 3** with the selected model and UFBoot (1,000 replicates). Alternatively leave the model on `AUTO` and let IQ-TREE run ModelFinder internally.

7. Open the tree in IcyTree. Use **Style → Internal node text → Label** to show UFBoot support. Reroot at the *Pseudiberus* clade (MN056414 and MN056415) by hovering the branch and Shift-clicking.

8. Export the tree as Newick/NEXUS for reuse or as SVG/PNG for a figure.

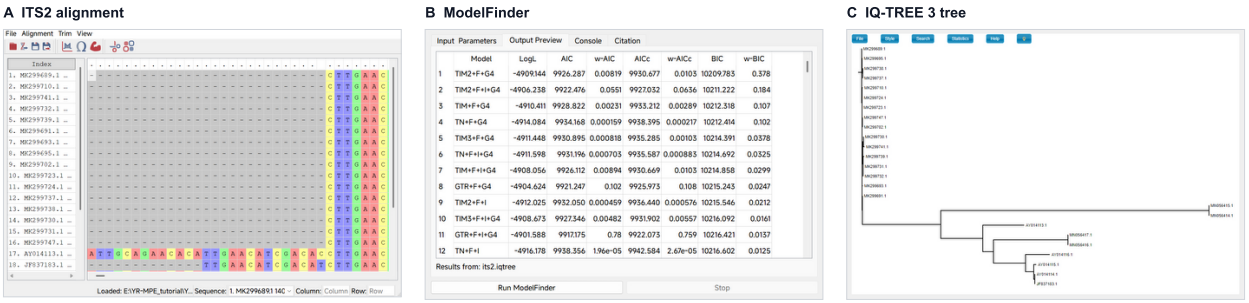


Figure 1.1: Single-gene ITS2 case. (A) The 25-sequence alignment after MAFFT with `--adjustdirection` and TrimAl. (B) ModelFinder BIC table; the selected model is TIM2+F+G4. (C) IQ-TREE 3 maximum-likelihood tree with UFBoot support, displayed in IcyTree.

1.4.1 Alternative ML engine: RAxML-NG

RAxML-NG is a second maximum-likelihood engine for the same single-gene alignment. It is not a separate case: use the same trimmed alignment and the same model-selection decision, then choose the engine that fits the analysis.

- Open **PHYLOGENY → Maximum Likelihood → RAxML-NG**. Load the alignment and choose a RAxML-NG-compatible model. Model names are not identical to IQ-TREE: a model selected as `TIM2+F+G4` in ModelFinder may need a compatible RAxML-NG model or a ModelTest-NG run.

- Set bootstraps with `--bs-trees` (1,000 for a final tree) and set threads explicitly. For partitions, provide a RAxML partition file or build the partitioned matrix in the workbench.

- Root the tree after inference at the outgroup; RAxML-NG returns an unrooted ML tree.

Use IQ-TREE when ModelFinder, partition merging and a single model-selection report are the priority. Use RAxML-NG when its model set or its throughput is preferred. Report the engine, model string and bootstrap method; do not mix support values from the two programs in one figure.

1.5 Parameter rationale

ITS2 is non-coding and length-variable, so a generic nucleotide aligner with direction correction is appropriate. `-automated1` removes gap-rich, low-information columns without manual, non-reproducible trimming. BIC is the default ModelFinder criterion and penalises the extra parameters of the TIM2+F+G4 model. UFBoot is fast but should use at least 1,000 replicates for a final tree.

1.6 Expected output

A 25-taxon alignment, a ModelFinder table, an IQ-TREE report and a Newick tree. The tree should contain all 25 accessions; the outgroup clade MN056414/MN056415 is used only for display rooting.

1.7 Interpretation

The tree shows the ITS2 relationships recovered under the selected model. Support values are UFBoot percentages, not posterior probabilities. Because ITS2 is a single non-coding marker, the topology is a hypothesis for the sampled region, not a species tree; concordant nuclear and mitochondrial markers are needed before taxonomic conclusions.

1.8 Pitfalls

Do not align ITS2 with MACSE. Check for reverse-complemented records before trusting a gap-rich alignment; `--adjustdirection` is the fast fix. Do not over-trim: removing too many variable columns lowers likelihood and can collapse the ingroup. Rooting is a display operation for time-reversible models; it does not change the unrooted likelihood.

1.9 Citation

MAFFT [1]; TrimAl [2]; IQ-TREE 3 [3]; IcyTree (Vaughan, 2017); the ITS2 data are from Wu and Liu (2019).

Chapter 2

Concatenated multi-gene phylogeny from Sanger amplicons

2.1 Question

Can a partitioned alignment assembled from raw Sanger reads recover the species tree when one locus is incomplete and some reads have an unknown orientation?

2.2 Data

The example uses `benchmark/open_tasks/task02`: 12 taxa, three amplicons spanning the ribosomal cluster (18S, ITS and 28S), and one missing ITS record (`B3421`). The FASTAs are unaligned amplicon exports; the ITS partition contains high-indel regions.

2.3 Before you start

Open a persistent workspace (**WORKSPACE** → **Select Workspace** → **Select Folder**), because the case keeps the per-locus alignments and the concatenated matrix as separate workspace items.

2.4 Steps

1. Open **WORKSPACE** → **Dataset** → **Create by SeqMatrix**. Import the three FASTA files with **Import from Sequences**. Drag each file's sequences into the corresponding partition column and rename the columns `18S`, `ITS` and `28S`. The empty `B3421` ITS cell is left blank; SeqMatrix will encode it as `?` in the supermatrix.
2. Click **Import as a dataset** to send the matrix and its partition definitions to the workbench.
3. Align each partition with **ALIGN** → **Align by** → **MAFFT**. In **Advanced Parameters**, set **Direction adjustment** to **Adjust direction** (`--adjustdirection`). This detects reads

that were returned in the reverse orientation and reverse-complements them before alignment. Keep the option off for protein data.

4. Trim the ITS alignment with **ALIGN** → **Trim Alignment by** → **TrimAl** using `-automated1`, and inspect the trimming report before accepting it.

5. Open **Dataset** → **Sequence Concatenator**, add the three aligned partitions, and export the concatenated FASTA plus a partitioned NEXUS charset file. Confirm that the concatenation order and the taxon headers match across partitions.

6. Run **MODEL** → **Find Best Substitution Model (ML)** in partition mode with edge-linked proportional branch lengths, then **PHYLOGENY** → **Maximum Likelihood** → **IQ-TREE 3** with the resulting partition scheme and UFBoot.

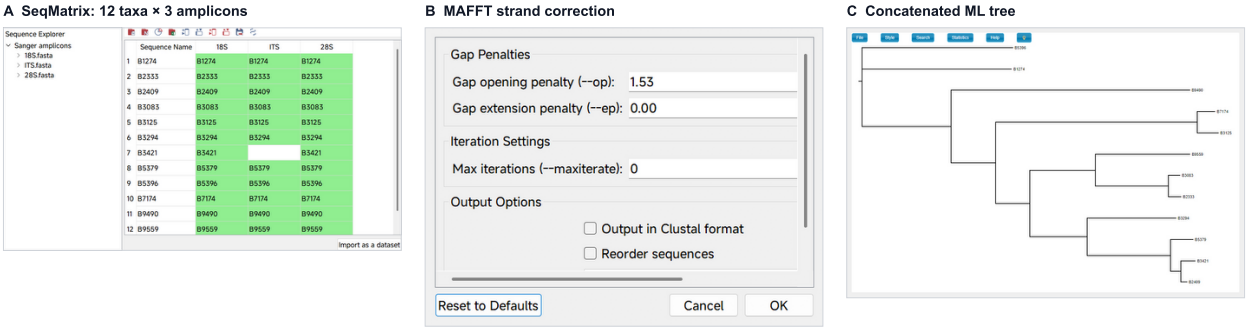


Figure 2.1: Multi-locus Sanger case. (A) SeqMatrix with 12 taxa and the 18S, ITS and 28S partitions; the missing ITS amplicon is retained as an empty cell. (B) MAFFT advanced parameters with `--adjustdirection` selected. (C) Concatenated maximum-likelihood tree inferred with IQ-TREE 3 and displayed in IcyTree.

2.5 Parameter rationale

`--adjustdirection` is appropriate because Sanger reads are often assembled without orientation control; the option is limited to nucleotide data and requires at least two sequences. Partitioned model selection is used because the three amplicons differ in substitution process and gap distribution. `-automated1` removes the most aggressively gapped ITS columns while retaining parsimony-informative sites.

2.6 Expected output

A 12-taxon concatenated alignment, a partition file, an IQ-TREE report and a rooted-for-display Newick tree. The tree should contain all 12 taxon labels; the missing ITS record must not appear as an empty taxon.

2.7 Interpretation

The case demonstrates the intended handoff: unaligned reads → partitioned matrix → per-locus alignments → trimmed supermatrix → model → tree. The topology should be read as a demonstration on a small simulated/benchmark data set, not as a taxonomic conclusion.

2.8 Pitfalls

Taxon matching in the concatenator uses the exact FASTA header. Rename headers before concatenation if the source files disagree. Do not concatenate unaligned amplicons. Check the `_R_` prefixes that MAFFT adds when it reverse-complements a read, and verify that the ITS alignment does not retain long gap-only blocks.

2.9 Citation

MAFFT [1]; TrimAl [2]; IQ-TREE 3 [3]; SeqMatrix is part of YR-MPE.

Chapter 3

Comparative organelle genomics

3.1 Question

How do gene order and annotation differ among annotated organelle genomes, and how can a standardized per-locus data set be exported for phylogeny?

3.2 Data

Four complete chloroplast genomes from the test data: *Eucalyptus grandis* (HM347959), *Sesamum indicum* (JN637766), *Andrographis paniculata* (KF150644) and *Camellia pubicosta* (KJ806277). Each GenBank record contains gene, CDS, tRNA and rRNA features.

3.3 Before you start

Keep the GenBank files as one folder. SeqDBG reads the feature table and does not need an external alignment program.

3.4 Steps

1. Open **WORKSPACE** → **Dataset** → **Create by SeqDBG**. Add the four `.gb` files with the + button.
2. In **Taxon Management**, confirm that all four genome identifiers are present. In the **Gene Statistics** table, inspect the left/right neighbour counts; a red neighbour count marks a rearrangement or an annotation inconsistency.
3. Click a gene row (for example `ycf15`) to draw its left–current–right adjacency Sankey plot on the **Plot** tab.
4. Review the **Standardization** tab. The built-in rules map common aliases such as `cox1` to `COX1`; add project-specific rules if the annotations use different names for the same gene.

5. On the **Dataset** tab choose **Nucleotide** (CDS, tRNA, rRNA) or **Amino Acid** (CDS), select all loci or a custom set, and export FASTA files or send the dataset to the workbench.

6. Align the CDS partitions with MACSE 2 (or MAFFT for non-coding partitions), trim, concatenate, and infer the species tree with IQ-TREE 3.

3.4.1 Alternative path: One-Step OrthoFinder

SeqDBG is the right tool when the annotations share a usable gene vocabulary and the question is about gene order or annotation consistency. When the input is a batch of genomes with heterogeneous or anonymous annotations, use the **One-Step OrthoFinder for Organelle Genomes** plugin instead.

1. Open **WORKSPACE** → **Dataset** → **One-Step OrthoFinder**. Select a folder of GBFF/GB/GBK files, one file per organelle genome, or a multi-record GenBank file. Set the genetic code, pseudogene handling and internal-stop policy.

2. The plugin parses the CDS features, translates them, and writes `proteomes/<sample>.faa`, `cds/<sample>.fna` and `cds_map.tsv`. It then runs OrthoFinder and identifies single-copy orthogroups.

3. In the output table, inspect the single-copy orthogroups and the species/CDS/OG summary. For each orthogroup the plugin writes a protein FASTA, an in-frame CDS FASTA and a mapping table.

4. Send the CDS orthogroups to MACSE 2 for codon-aware alignment, trim with TrimAl, concatenate, and infer the tree with IQ-TREE 3 as above.

The two routes answer different questions. SeqDBG visualises and exports *annotated features*; OrthoFinder infers *orthology from sequence similarity* and is therefore better when gene names are inconsistent or when the annotations contain duplicated copies. Use SeqDBG first to understand the gene-order variation, then OrthoFinder for the orthology-based matrix.

3.5 Parameter rationale

SeqDBG is used before alignment because organelle annotations frequently differ in gene naming and boundaries. Standardizing names and inspecting adjacency first prevents non-orthologous CDS sequences from entering the concatenation. MACSE is preferred for CDS because it aligns at the codon level and flags frameshifts and premature stops.

3.6 Expected output

A standardized per-locus FASTA set with one sequence per genome and locus, a gene inventory, and a species tree. The SeqDBG table should show the selected gene in all four genomes; missing genes are visible as reduced taxon counts.

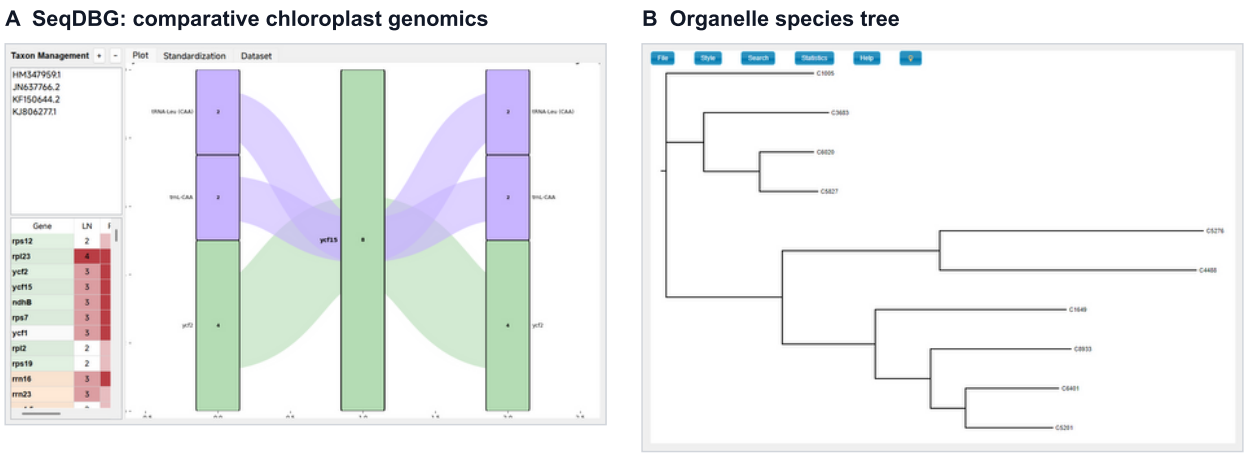


Figure 3.1: Comparative organelle case. (A) SeqDBG gene-statistics table and adjacency Sankey for four annotated chloroplast genomes; the selected gene and its left/right neighbours are highlighted. (B) Species tree recovered from the linked organelle CDS workflow and displayed in IcyTree.

3.7 Interpretation

Conserved adjacencies support collinearity; red or asymmetric neighbour counts flag rearrangements, duplicated annotations or naming differences. The tree demonstrates the workflow, not the organismal phylogeny.

3.8 Pitfalls

Amino-acid export requires a `/translation` qualifier on every CDS; verify this before export. Nucleotide export skips records whose identifier contains `_protein`. Do not treat a whole organelle contig as one locus; the point of the case is per-feature orthology.

3.9 Citation

SeqDBG and YR-MPE; MACSE 2 [4]; IQ-TREE 3 [3].

Chapter 4

Distance-based phylogeny

4.1 Question

When is a distance tree an appropriate first look at a data set, and how do p-distances, model-corrected distances and substitution saturation affect the result?

4.2 Data

`examples/cases/distance/` contains the trimmed ITS2 alignment from Chapter 1 and two distance matrices: the IQ-TREE `.mldist` file and an eight-taxon square subset for display. The alignment is 25 taxa × 1,408 columns; the subset keeps five ingroup accessions, *Vitrea* and the two *Pseudiberus* outgroup sequences.

4.3 Before you start

Use a persistent workspace. Distance methods need a rectangular alignment; remove or mask columns with excessive ambiguity first. Decide whether gaps are handled by pairwise or complete deletion and keep that choice fixed across the comparison.

4.4 Steps

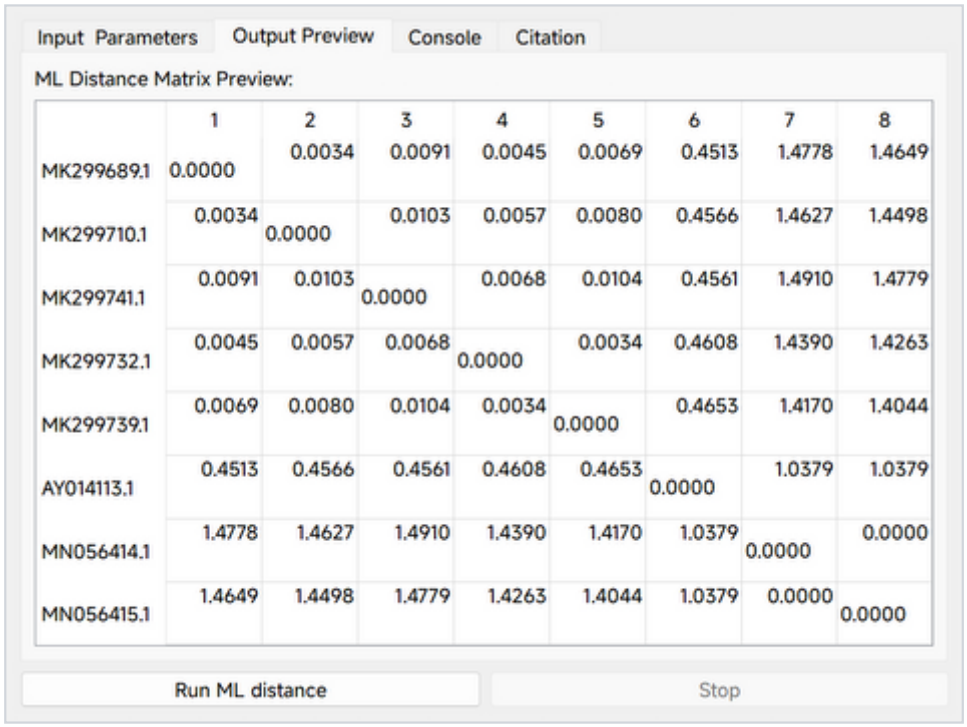
1. Import `its2_trimmed.fasta` with **ALIGN → Open Sequence Files**.
2. Run **DISTANCE → Compute Uncorrected Distances (p-distance)** with pairwise deletion. Inspect the matrix; p-distance is the raw proportion of differing sites and does not correct multiple hits.
3. Run **DISTANCE → Compute Pairwise Distances (ML)** with IQ-TREE 3, model `TIM2+F+G4` and the same gap policy. The plugin writes a `.mldist` matrix and previews it.

4. Run **DISTANCE → Calculate & Plot Substitution Saturation** with pairwise deletion. Read the Ti/Tv versus p-distance plot and the C-value; a strongly saturating marker should not be used for deep nodes.

5. Open **PHYLOGENY → Distance Methods → DecentTree**. Select the ML distance matrix and choose `BIONJ-R` (or another NJ variant).

6. Inspect the tree in IcyTree. If the bundled DecentTree build fails on a particular matrix, use the IQ-TREE NJ tree from the same matrix as a fallback and record which program produced the figure.

A ITS2 ML distance matrix



	1	2	3	4	5	6	7	8
MK299689.1	0.0000	0.0034	0.0091	0.0045	0.0069	0.4513	1.4778	1.4649
MK299710.1	0.0034	0.0000	0.0103	0.0057	0.0080	0.4566	1.4627	1.4498
MK299741.1	0.0091	0.0103	0.0000	0.0068	0.0104	0.4561	1.4910	1.4779
MK299732.1	0.0045	0.0057	0.0068	0.0000	0.0034	0.4608	1.4390	1.4263
MK299739.1	0.0069	0.0080	0.0104	0.0034	0.0000	0.4653	1.4170	1.4044
AY014113.1	0.4513	0.4566	0.4561	0.4608	0.4653	0.0000	1.0379	1.0379
MN056414.1	1.4778	1.4627	1.4910	1.4390	1.4170	1.0379	0.0000	0.0000
MN056415.1	1.4649	1.4498	1.4779	1.4263	1.4044	1.0379	0.0000	0.0000

Figure 4.1: ITS2 distance case. The eight-taxon subset of the ML distance matrix under TIM2+F+G4. The full matrix is in `examples/cases/distance/its2.mldist`.

4.5 Parameter rationale

p-distance is fast and assumption-free but underestimates substitutions when multiple hits are common. ML distance uses the substitution model and is the better input for a distance tree when the model is adequate. Pairwise deletion retains more information in a gappy alignment; complete deletion is cleaner but can remove most of the variable sites. BIONJ is a fast neighbour-joining variant; it returns an unrooted tree.

4.6 Expected output

A p-distance matrix, an ML distance matrix, a saturation plot and a Newick tree. The tree may contain negative branch lengths. This is a property of fitting an additive tree to a non-additive distance matrix, not evidence of negative evolutionary time; do not pass such a tree to a dating program without correcting or constraining the branch lengths.

4.7 Interpretation

A distance tree is a summary of pairwise similarity, not a full likelihood analysis. It is useful for a quick check of taxon placement, for identifying obvious mislabels and for producing a starting tree. If the ML and p-distance trees disagree strongly, inspect the alignment and the saturation plot before choosing either.

4.8 Pitfalls

Do not compare distances computed under different gap policies. Do not use a saturated marker for a deep divergence. Do not interpret a negative branch as a biological result. If DecentTree rejects a lower-triangular `.mldist` file, convert it to a square matrix first; the example includes both forms.

4.9 Citation

IQ-TREE 3 [3]; DecentTree [5]; the ITS2 data are from Wu and Liu (2019).

Chapter 5

Bayesian inference with MrBayes

5.1 Question

How can a MrBayes analysis be configured with independent runs and multiple Metropolis-coupled chains, and how are its traces checked before the consensus tree is accepted?

5.2 Data

The example uses `dev/fig3/mrbayes/locus001.nex`, a 10-taxon nucleotide alignment with 1,000 sites. The two MrBayes runs in the same folder (`locus001.nex.run1.p` and `locus001.nex.run2.p`) are the input for the diagnostics.

5.3 Before you start

Select a persistent workspace. MrBayes writes checkpoint, `.p`, `.t` and consensus files during the run; a temporary workspace is not sufficient. Keep MPI and BEAGLE enabled only when the runtime check passes.

5.4 Steps

1. Open **PHYLOGENY → Bayesian Inference → MrBayes3-MPI-BEAGLE**. Load `locus001.nex`. The plugin reads the existing MrBayes block or converts the alignment to an interleaved NEXUS block.

2. Set **Type** to `DNA`, the substitution model to `GTR` (six rate classes) and state frequencies to `estimated(dirichlet)`. Keep **Gamma (+G)** with four categories.

3. In **MCMC & Run Settings**, set **Generations** to 30,000, **Sampling Freq.** to 100, **Runs** to 2 and **Chains** to 4. Set the burn-in fraction after inspecting the traces; do not guess it before the run.

4. Run MrBayes. When it finishes, open the consensus tree in IcyTree and confirm that the taxon set and support labels are present.

5. Open **PHYLOGENY → Bayesian Inference → MCMC Diagnostics (MiniTracer)**. Load both `.p` files and set the burn-in fraction (0.100 in the example).

6. Inspect the ESS column. Rows are green at ESS ≥ 200, yellow at 100–200 and red below 100. Select `lnLike` to see its post-burn-in trace, marginal density, 95% HPD and summary statistics.

7. Select both files and open **Compare 2 Chains**. The table reports each chain's sample count and mean, the correlation coefficient **R**, and the Gelman–Rubin **PSRF (R-hat)**. Values close to 1 indicate agreement between runs.

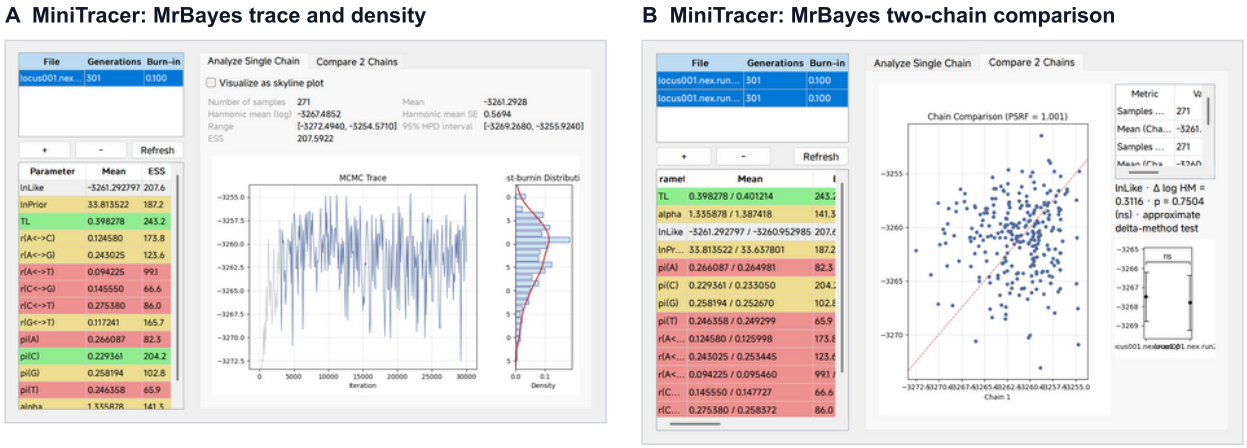


Figure 5.1: MrBayes convergence case. (A) MiniTracer single-chain view for `lnLike`: post-burn-in trace, marginal density, ESS and 95% HPD. (B) Two-run comparison with the Gelman–Rubin PSRF (R-hat) in the table and plot title.

5.5 Parameter rationale

Two independent runs are the minimum needed to separate within-chain from between-chain variance. Four chains per run allow Metropolis coupling to escape local optima. GTR + Γ_4 is a flexible nucleotide model; the gamma categories absorb rate variation among sites. MPI and BEAGLE accelerate the likelihood without changing the model.

5.6 Expected output

Two `.p` traces, two `.t` tree files, a consensus tree and MiniTracer diagnostics. A converged analysis has ESS ≥ 200 for every reported parameter, stable traces without trends, and PSRF close to 1.

5.7 Interpretation

ESS measures within-chain mixing; PSRF compares between-chain and within-chain variance. Neither statistic proves that the model is correct. The PSRF in MiniTracer is a per-parameter two-chain R-hat, not a multivariate PSRF over the whole model. Read it together with the trace, the density and the MrBayes `sumt` output.

5.8 Pitfalls

A short chain can look stationary while ESS is still far below 200. Burn-in is entered as a fraction, not a generation count. Do not compare runs with different parameter sets. MiniTracer truncates the two chains to the shorter post-burn-in length before computing PSRF, so unequal run lengths are handled conservatively.

5.9 Citation

MrBayes [6]; MiniTracer is part of YR-MPE.

Chapter 6

Inferring deep evolution with infinite mixture models

6.1 Question

How can a site-heterogeneous infinite-CAT model be used for a deep phylogenetic problem, and how is the resulting chain summarised?

6.2 Data

`dev/cases/bayes/prmA.phy` is a 55-taxon, 411-amino-acid alignment used to demonstrate the PhyloBayes-MPI infinite CAT configuration. The alignment is deliberately small enough to run on a workstation; a production analysis should use the full marker set and a much longer chain.

6.3 Before you start

Use a persistent workspace and check that the PhyloBayes-MPI and `mpiexec` paths resolve. CAT is computationally expensive; budget more time than for a nucleotide GTR analysis of the same size.

6.4 Steps

1. Open **PHYLOGENY** → **Bayesian Inference** → **PhyloBayes-MPI**. Load `prmA.phy` as a PHYLIP alignment.
2. Enable **Site Heterogeneity Model** and select **Infinite (CAT)**. Keep **Rate Heterogeneity** on with four gamma categories and choose **GTR** for the exchangeabilities.
3. Set **Number of Parallel Chains** to 2 or more and set a chain length appropriate for the data. For a real analysis, increase the generations until the traces and `bpcomp` agree; do not stop at the short demonstration length.

4. Run PhyloBayes-MPI. When the chains finish, open **Compare & Consensus**, load the chains from the results, and run **bpcomp** to obtain the majority-rule consensus and the bipartition list.

5. Treat `maxdiff` below 0.1 as agreement between chains. Inspect the consensus tree in IcyTree and export it as Newick/NEXUS.

6. If the chains disagree, extend the run, increase the burn-in, or add chains; do not average chains that have not converged.

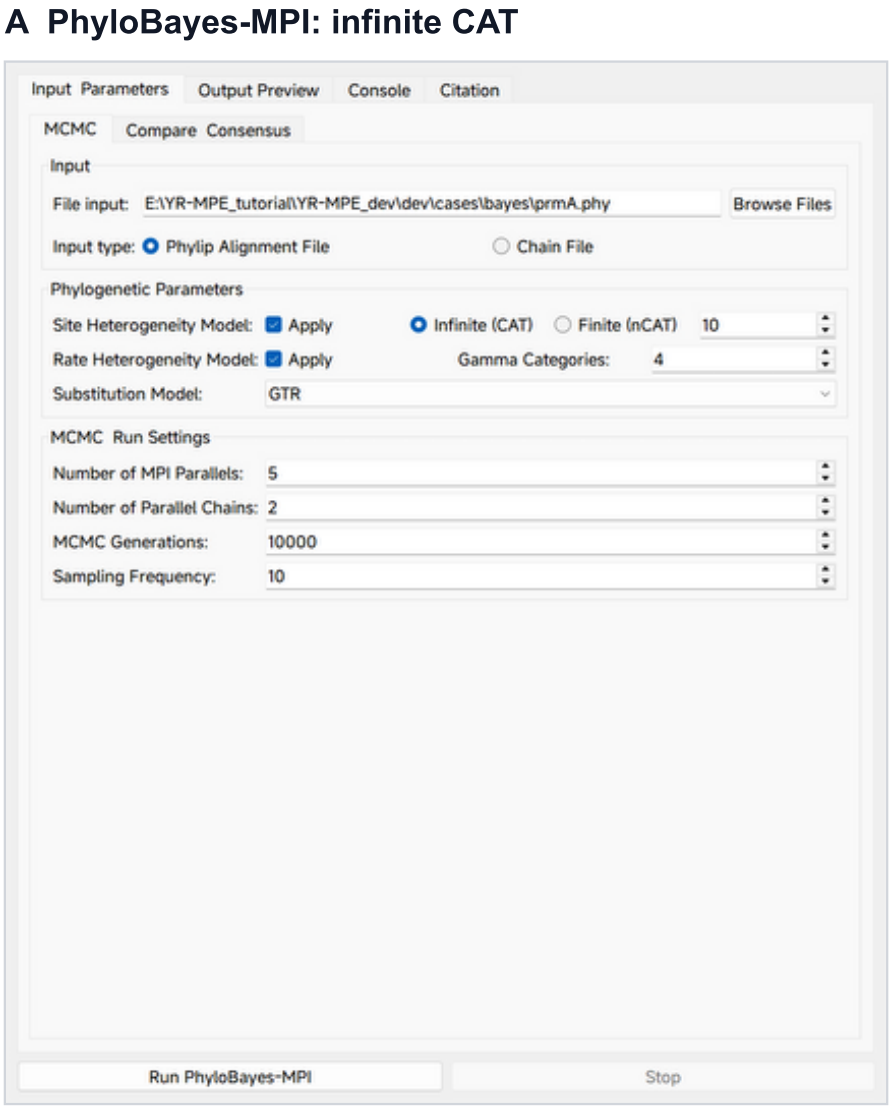


Figure 6.1: PhyloBayes-MPI configured with GTR, four discrete gamma categories and the infinite CAT mixture. Two independent chains are requested; the chain length shown is a demonstration value, not a convergence guarantee.

6.5 Parameter rationale

The infinite CAT mixture lets each site draw from a potentially unbounded number of equilibrium frequency profiles. This is useful when the data are site-heterogeneous and long-branch attraction is a concern. GTR supplies the exchangeability matrix; the discrete gamma handles rate variation. CAT is a mixture of profiles, not a partition model, so it does not require a partition file.

6.6 Expected output

Two or more PhyloBayes chains, a consensus tree, a bipartition list and a `bpcomp maxdiff` close to zero. A converged analysis has stable traces and agreement between independent chains.

6.7 Interpretation

The CAT model is a statement about site heterogeneity, not a guarantee that the topology is correct. Compare the result with a homogeneous model and with a partition model; a topology that appears only under CAT is a hypothesis that needs independent support. Read `maxdiff` together with the trace plots, not as a substitute for them.

6.8 Pitfalls

CAT is slow; a short chain may look stationary while the tail is still moving. Do not confuse infinite CAT with finite nCAT: nCAT fixes the maximum number of profiles. Do not compare chains with different parameter sets or different taxon sets.

6.9 Citation

PhyloBayes [7]; the CAT model follows Lartillot and Philippe (2004).

Chapter 7

Species tree, concordance factors, and reconciliation

7.1 Question

How do gene-tree and site-based concordance factors localise conflict on a species tree, and how is a gene tree reconciled with that species tree?

7.2 Data

`examples/cases/species_tree/` contains a simulated 10-taxon species tree (`species_ils.nwk`), 150 gene trees simulated under incomplete lineage sorting (`gene_trees_150.trees`), and the gCF, sCF and MSC diagnostic tables. A duplicated gene tree (`dup_gene.nwk`) is included for the TreeRecs exercise.

7.3 Before you start

All gene trees must use the same taxon labels as the species tree. ASTRAL treats gene trees as unrooted; the species tree is used as a fixed tree for gCF and sCF. If wASTRAL is requested, the gene-tree node labels must be numeric; trees without numeric labels are rejected rather than silently downweighted.

7.4 Steps

1. Open **SPECIES TREE → ASTRAL-IV** and load the 150 gene trees. Run ASTRAL-IV, then repeat with wASTRAL if the gene trees have numeric support labels. ASTRAL-Hybrid is a wASTRAL preset, not a third method.
2. Open **SPECIES TREE → Gene Concordance (gCF)** with the species tree and the same gene trees. Inspect the gCF table and the split names; every split is named by its smaller taxon side.

3. Open **SPECIES TREE → Site Concordance (sCF)** with the species tree and the alignment used for the gene trees. Use the likelihood estimator and keep the quartet count fixed.

4. Open **SPECIES TREE → MSC Diagnostics (gCF + ASTER CU)**. Read the Table, Overview, Simplex and Bipartition info tabs together. Each simplex point is an internal split; a 10-taxon tree has at most seven points.

5. Open **SPECIES TREE → TreeRecs 1.2** with `dup_gene.nwk` and the species tree. Inspect the reconciliation SVG; squares mark duplications and crosses mark losses. Export the Newick event tree if needed.

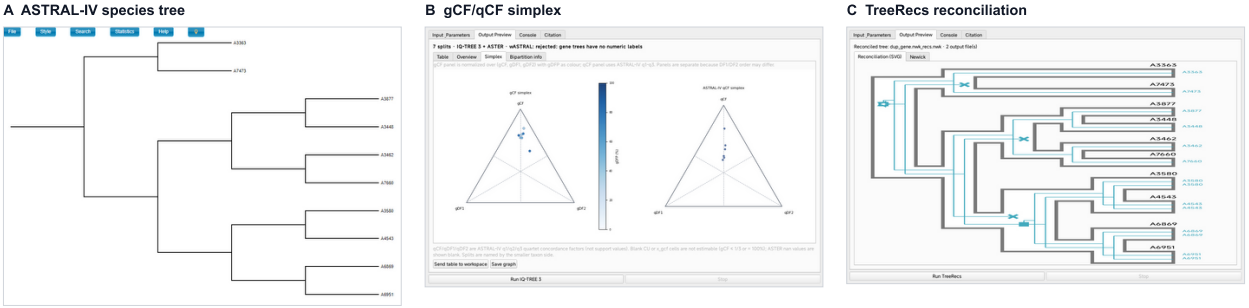


Figure 7.1: Species-tree case. (A) ASTRAL-IV species tree from the 150-gene-tree ILS data set. (B) gCF/qCF simplex; each point is an internal split and colour encodes gDFP. (C) TreeRecs reconciliation of a duplicated gene tree, with duplication and loss symbols.

7.5 Parameter rationale

ASTRAL-IV is the standard unweighted quartet method. wASTRAL downweights poorly supported quartets and long branches; ASTRAL-Hybrid is its support/length preset. gCF measures the proportion of decisive gene trees supporting a split; sCF measures the same idea at the site level. They answer different questions and should not be treated as branch support. TreeRecs is used after the species tree is accepted; it explains a gene tree, not the species tree itself.

7.6 Expected output

A species tree, gCF/sCF tables, a simplex view, a bipartition matrix and a reconciliation with duplication/loss events. The tables must use the same split names as the species tree. ASTRAL/wASTRAL CU cells are blank when the gCF is at or below 1/3 or exactly 100%; those values are not estimable.

7.7 Interpretation

A split near the gCF vertex is concordant; a split near a gDF vertex is in conflict. A high gDFP colour value means polyphyly or other non-NNI conflict, not a clean alternative topology. ASTRAL-

IV qCF is a quartet concordance factor, not a support value. The TreeRecs events are hypotheses under the chosen gene/species tree pair.

7.8 Pitfalls

Do not read the simplex as a posterior distribution; it has one point per split, not one point per gene tree. Do not compare simplexes from different taxon sets. Do not force a CU inversion when gCF $\leq 1/3$. A wASTRAL run without numeric labels will be rejected; use ASTRAL-IV or add valid labels.

7.9 Citation

ASTRAL/ASTER [8]; wASTRAL (Zhang and Mirarab, 2022); IQ-TREE 3 [3]; TreeRecs is described in the YR-MPE plugin documentation.

Chapter 8

Molecular dating

8.1 Question

How can a fixed phylogeny be converted into a time-scaled tree with a root calibration, and what is the difference between least-squares dating and a Bayesian clock analysis?

8.2 Data

`examples/cases/dating/` contains a simulated six-taxon tree (`simulated6.nwk`) with branch lengths in substitutions and a date file (`simulated6.dates`) that sets all tips to 0 and the root to −60 Mya. The expected output is `simulated6_dated.nexus`, a NEXUS tree with `[&date="..."]` annotations on every node.

8.3 Before you start

Use a persistent workspace. Decide the time unit once (years, Mya or million years) and keep it for every calibration. Do not date a tree with negative or zero branches without inspecting it; LSD2 collapses very short branches by default. If dates are years before present, the root age is negative and the tips are 0; the opposite sign produces a temporal-constraint conflict.

8.4 Steps

1. Import `simulated6.nwk` with **ALIGN** → **Open Sequence Files**. Root the tree at the outgroup or let LSD2 estimate the root.
2. Open **CLOCK** → **LSD2**. Load the tree and set the sequence length to 1,000.
3. Enable **Use tip dates** and set the tips date to 0. Add a root calibration: **Calibration points** → **+** → **Edit**, select all six taxa, choose a fixed date and enter −60.

4. Optionally enable **Calculate confidence intervals** with the default lognormal standard deviation $q = 0.2$. The interval is generated by simulating branch-length noise; it is not a posterior distribution.

5. Run LSD2. Inspect the dated NEXUS tree and the rate estimate in the report. The example gives a rate of about 3.30×10^{-3} substitutions per site per time unit and a root date of -60.

6. For a full Bayesian alternative, open **CLOCK** → **MrBayes3-MPI-BEAGLE**, enable **Phylogenetic Dating**, choose a strict or relaxed clock, add the same calibration and run the chain.

8.5 Parameter rationale

LSD2 is a least-squares method: it finds a rate and node dates that minimise the squared deviation from the observed branch lengths. It is fast and needs only a fixed tree and a few calibrations. MrBayes co-estimates topology and dates and returns credible intervals, at the cost of a much longer MCMC run. Use a strict clock when the rate is expected to be constant; use a relaxed clock (IGR, CPP or TK02) when branch-rate variation is plausible.

8.6 Expected output

A time-scaled tree with node dates, a rate estimate and, if requested, confidence intervals. The date annotations should increase from the tips (0) to the root (-60). A root date that is younger than a tip, or a calibration that conflicts with another calibration, produces an error rather than a dated tree.

8.7 Interpretation

The dated tree is a hypothesis conditional on the tree, the calibrations and the clock model. LSD2's intervals are simulation-based and should not be read as Bayesian credible intervals. MrBayes intervals are posterior summaries but still depend on the priors and the calibrations.

8.8 Pitfalls

Do not mix time units. Do not pass a distance tree with negative branches to a dating program. Do not use a point calibration when the fossil only provides a minimum bound. Do not compare rates from analyses with different time units or different root calibrations.

8.9 Citation

LSD2 [9]; MrBayes [6]; the example tree is simulated for this manual.

Chapter 9

Selection analysis

9.1 Question

How can a codon alignment be screened for departures from neutrality with pairwise d_N/d_S estimates, and when is a codon model preferable?

9.2 Data

`examples/cases/selection/codon.fas` contains four short codon alignments (A–D), 26 codons each. The example is deliberately small: it demonstrates the table and matrix interface, not a biological conclusion. The same alignment is used for NG86, YN00 and the batch MLE route.

9.3 Before you start

The input must be an in-frame codon alignment (length a multiple of three) with no internal stop codons in the analysed codons. Check the reading frame and remove alignment columns that are mostly gaps or ambiguous bases. For a real analysis, use a codon-aware alignment (MACSE 2) rather than a nucleotide alignment translated after the fact.

9.4 Steps

1. Open **SELECTION → Codon-based Z-test**. Load `codon.fas` and confirm the reported number of usable codons.
2. Choose **NG86** and run. The table reports S, N, S_d, N_d, d_S, d_N, $\omega = d_N/d_S$, Z and a conclusion for every sequence pair.
3. Switch the view to **Matrix (MEGA style)**. The upper triangle contains two-sided p-values; the lower triangle contains ω. Significant cells are bold and tinted: red for $d_N > d_S$, green for $d_N < d_S$.

4. Repeat with **YN00**. The F84 correction and its standard errors can change the call; compare the two estimators rather than reporting only one.

5. For a larger set, open **SELECTION → Batch dN/dS**. Use *Counting* mode for NG86/YN00 or *MLE* mode for IQ-TREE `--seqtype CODON` with `-m GY` or `-m MG`.

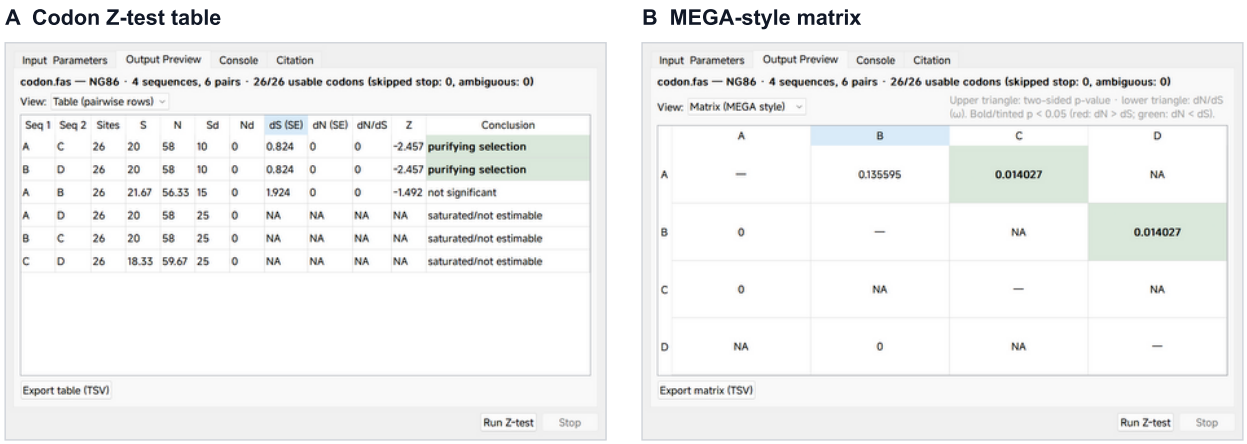


Figure 9.1: Codon selection case. (A) Pairwise NG86 table with d_S, d_N, ω, Z and the selection call. (B) MEGA-style matrix; upper triangle is the two-sided p-value and lower triangle is ω.

9.5 Parameter rationale

NG86 is a fast counting method with a Jukes–Cantor correction; YN00 adds an F84 correction and estimates transition/transversion bias. Both are pairwise and assume a common codon frequency model. The MLE route fits a codon substitution model to the whole alignment and is preferable when the alignment is long enough and the phylogeny is known or co-estimated.

9.6 Expected output

A pairwise table, a matrix view and, in batch mode, one row per pair or alignment. A d_S that is saturated or not estimable is reported as `NA`, not as zero. A significant negative Z with $d_N < d_S$ indicates purifying selection; a significant positive Z with $d_N > d_S$ indicates positive selection; a non-significant test is not evidence of neutrality.

9.7 Interpretation

The example's very short alignment produces unstable d_S estimates and several saturated pairs. That is the intended lesson: a small p-value from a short alignment is not a robust selection claim. Use a codon-aware alignment, check the d_S standard error and compare NG86, YN00 and MLE before interpreting a single pair.

9.8 Pitfalls

Do not translate a nucleotide alignment in a different frame. Do not treat `NA` as zero. Do not compare ω across pairs with very different d_S precision. CodeML/HyPhy site and branch-site models are not exposed in this version; the menu entry is a placeholder.

9.9 Citation

NG86 and YN00 follow the methods implemented in PAML [10]; the MLE route uses IQ-TREE 3 [3].

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